2. (**6 Points**) Give the C code instruction sequence to enable external interrupt 0 (INT1) as an interrupt source and to make it falling-edge triggered. What signal (port and port pin) must be low to trigger a level-low event for INT1?

PD3

```
LDI R16, 0B10000000
OUT    GIFR, R16
LDI R16, 0B00000000
OUT MCUCR, R16
CBI DDRD, 3; PD3
SBI PORTD, 3; PD3 INPUT FOR INT1
SEI
```

```
P.
DDRD &= 0b11110111;
PORTD |= 0b00001000;
MCUCR = 0b00001000;
GICR = 0b10000000;
sei();
```

3. Interrupt priority.

Suppose that the interrupt sources Timer 2 CTC (Clear Timer on Compare Match), Timer 0 timer overflow (TOV0), external interrupt 1 (INT1), and the receive complete serial interrupt are enabled (ATmega32). Answer the following questions.

a) **(9 Points)** What are the interrupt flags that are used/monitored for each of the four interrupt sources to determine if it (interrupt source) needs to be serviced? What I/O register(s) are the interrupt flags contained? For example, the interrupt flag for Timer0 overflow (OVR) is TOV0.

I/O Reg | Flag

TIFR | Timer2 CTC : OCF2

TIFR | Timer0 TOV0 : TOV0

GIFR | external interrupt1 (INT1) : INTF1

UCSRA | receive complete serial interrupt : RXC

b) **(4 Points)** What is the order that these interrupt sources are serviced if they occur at the same time?

ORDER IN THE INTERRUPT VECTOR TABLE (LOWEST TO HIGHEST)

INTF1 — 0x0004

OCF2 — 0x0008

TOV0 — 0x0016

RXC — 0x001A

7. Given the C program below. Assume fosc = 8MHz.

(12 Points) Convert the assembly program (below) to **your own** C code (ATmega32) in the space on the left.

```
.ORG 0
RJMP MAIN

.ORG 0X0016
RJMP TIMER0_OVR_ISR

.ORG 50
LDI R16, 0X01
IN R17, DDRB
OR R17,R16
OUT DDRB, R17
LDi R16, 0XFF
OUT DDRC, R16
LDI R16, 0X00
OUT DDRD, R16
LDI R16, 0XFF
OUT PORTD, R16

LDI R16, -20
OUT TCNT0, R16
LDI R16, 0X02
OUT TCCR0, R16
LDI R16, 0X01
OUT TIMSK, R16
SEI
HERE: IN R18, PIND
OUT PORTC, R18
RJMP HERE

.ORG 100
TIMER0_OVR_ISR: IN R19, 0X01
IN R20, PORTB
EOR R20,R19
OUT PORTB, R20
LDI R16, -20
OUT TCNT0, R16
RETI
```

Put your C program code conversion from assembly program (shown on left) here.

```
#include <avr/io.h>
#include <avr/interrupt.h>
ISR(TIMER0_OVR_ISR);

Void main() {
    DDRB |= 0x01;
    DDRC = 0xFF;
    DDRD = 0x00;
    PORTD = 0xFF;
    TCNT0 = -20;
    TCCR0 = 0x02;
    TIMSK = 0x01;
    sei();
    while(1) {
        PORTC = PIND;
    }
}
ISR(TIMER0_OVR_ISR){
    PORTB ^= 0x01;
    TCNT0 = -20;
}
```